FX exposure analytics

Task 2 | SQL calculations with independent Python reconciliation

Prepared 14 October 2026 | Retrospective report on the saved project package

Finding: EUR drove the weekly translation decline

For hypothetical unhedged balances of EUR 10 million, GBP 5 million and JPY 1 billion, the combined USD value fell by $89,710 between 2 and 9 October 2026, a 0.371% decline. EUR contributed $66,000 of the decline. The latest combined value was $24,123,024.

Currency	5-observation return	FX-only USD change
EUR	-0.586%	-$66,000
GBP	-0.038%	-$2,500
JPY	-0.335%	-$21,210

The market observations are public data; the balances are invented teaching assumptions. Results measure the effect of spot FX translation on constant cash balances. They are not actual client performance, realized trading profit or accounting P&L.

Horizon to 9 Oct	Portfolio FX change	Portfolio FX return
1 observation (8 Oct)	+$9,601	+0.040%
5 observations (2 Oct)	-$89,710	-0.371%
21 observations (10 Sep)	-$762,954	-3.066%

The three horizons overlap and must not be summed. The largest percentage change and the largest dollar contribution need not belong to the same currency because the USD exposure sizes differ.

1. Quote conventions and valuation

FRED series	Raw units	Conversion to USD per currency
DEXUSEU	USD per EUR	Keep raw quote
DEXUSUK	USD per GBP	Keep raw quote
DEXJPUS	JPY per USD	Use 1 / raw quote

Quote direction is the critical control. If USD/JPY rises from 150 to 160, a fixed JPY 1 billion asset falls from $6.667 million to $6.250 million. The yen depreciates in USD terms even though the raw quote rises. The reciprocal return is nonlinear; it is not obtained by simply negating the original percentage change.

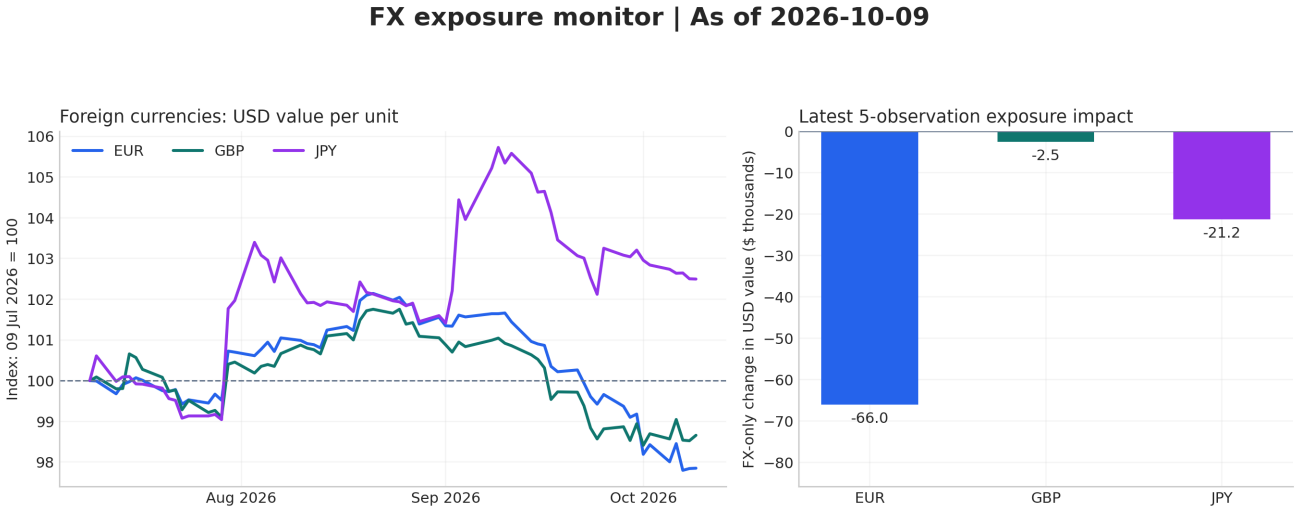
For foreign-currency amount Q and normalized rate r: USD value = Q x r. FX-only USD change = Q x (current r - prior r). Currency return = 100 x (current r / prior r - 1). The portfolio return is total dollar change divided by total starting USD value, not a simple average of currency returns.

SQL responsibilities

The raw_fx table retains original observations, the currencies table records quote conventions and the exposures table stores one fixed balance per currency. A common complete-date set aligns all comparisons. CASE normalizes direction; LAG calculates prior rates; joins attach exposures; aggregation produces the portfolio result.

LAG(usd_per_fcy, 5) OVER (PARTITION BY currency ORDER BY date) means: within one currency's time series, retrieve the rate five rows earlier. Calculate the window before filtering to the latest date, or the required history disappears. A unique exposure key prevents a join from multiplying balances.

2. Visual evidence and commercial use



Illustrative fixed holdings: EUR 10m | GBP 5m | JPY 1bn. Unhedged; interest and costs excluded.
Source: Federal Reserve H.10 via FRED. Common complete dates; no forward-fill. Reference rates, not live execution quotes.

The indexed panel starts each normalized currency at 100 on the first common date in the recent three-month window. An index of 102 means a 2% increase in USD value per currency unit from that starting point. The contribution panel translates rate changes into dollars for the stated balances.

Use the attribution to ask focused questions: are these assets, liabilities or future cash flows; what are the actual balances and settlement dates; what is the reporting currency; and how much is hedged? A spot monitor cannot price a forward hedge without additional contract and market inputs.

The Federal Reserve H.10 observations are reference rates, not live executable bank quotes. This is a descriptive client-analytics prototype rather than a hedge optimizer or intraday execution system.

3. Controls, reproduction and sources

Control	Purpose / recorded outcome
Unique date and currency keys	Prevent duplicate market observations and duplicated exposure joins.
Positive finite quotes	Reject zero, negative and infinite rates before inversion or valuation.
Common complete dates	Keep horizon endpoints consistent across currencies; no forward fill.
Python versus SQL	Original learning guide records independent normalization and reconciliation of all nine latest currency/horizon results, including dates and dollar changes.
Value identity	Current value minus prior value equals the FX attribution.

Remaining limits: holdings are constant, with no flows, interest, changes in non-cash asset prices, hedges or transaction costs. Missing dates are not fully reconciled against source calendars. The original guide records endpoint checks against FRED, but no second-publisher reconciliation of the full historical snapshot.

Extract BNY_Task2_FX_SQL_Pack.zip. In bny_task2 run python -m pip install pandas numpy matplotlib, then python task2.py. The included snapshot supports offline analysis after dependency installation. Open outputs/fx_monitor.sqlite in a SQLite viewer; inspect sql/schema.sql, analysis.sql and client_monitor.sql. The notebook provides a guided alternative.

AI-assisted development: the package implements real SQL windows, joins and valuation, with Python used for independent arithmetic checks and charting. Describe this as assisted development with review, not evidence of production-scale engineering or independent mastery merely because the script runs.

Report evidence: LEARNING_GUIDE.md, client_insight_memo.md, portfolio_summary.csv and the saved chart. The full pipeline was not rerun for this report. References: https://fred.stlouisfed.org/series/DEXUSEU ; https://fred.stlouisfed.org/series/DEXUSUK ; https://fred.stlouisfed.org/series/DEXJPUS ; https://www.sqlite.org/windowfunctions.html .